

Student Name: Priyanka Nair

Roll No: 12CHE003

Subject: Chemistry

School / Institution:	St. Xavier's Senior Secondary, Mumbai
Student Name:	Priyanka Nair
Roll Number:	12CHE003
Class:	XII-C
Subject:	Chemistry
Barcode:	BAR103
Date of Examination:	15 March 2025
Session:	2024-25

I hereby declare that the answers written in this booklet are my own. I have not used any unfair means during the examination.

Candidate's Signature: _____ Invigilator's Signature: _____ Room No: _____

SECTION A — Very Short Answer (1 mark each)**Q1.** What is Raoult's law?

[1 mark]

Raoult's Law: The partial vapour pressure of any volatile component of a solution is directly proportional to its mole fraction.

$p_A = x_A \times p_A^\circ$ where p_A° is vapour pressure of pure A and x_A is mole fraction.

Total pressure for ideal solution: $P = x_A p_A^\circ + x_B p_B^\circ$

Marks	
Awarded	
/1	
Initials	

Q2. Define electrochemical equivalent.

[1 mark]

Electrochemical equivalent (z) is the mass of a substance deposited or liberated at an electrode when 1 coulomb of charge is passed through the electrolyte.

Unit: g/C or kg/C

By Faraday's first law: $m = zQ = zIt$

Student Name: Priyanka Nair

Roll No: 12CHE003

Subject: Chemistry

Marks	
Awarded	
/1	
Initials	

Q3. What is a peptide bond?

[1 mark]

A peptide bond is an amide linkage formed between the carboxyl group (-COOH) of one amino acid and the amino group (-NH₂) of another amino acid with the elimination of a water molecule.

-CO-NH- is the peptide bond. Proteins are made up of amino acids linked by peptide bonds.

Marks	
Awarded	
/1	
Initials	

Q4. Give one example of a nucleophilic substitution reaction.

[1 mark]

SN₂ reaction (Bimolecular Nucleophilic Substitution):



The hydroxide ion (nucleophile) attacks from the back side, displacing Br⁻ (leaving group) in a single step with inversion of configuration (Walden inversion).

Marks	
Awarded	
/1	
Initials	

SECTION B — Short Answer (2 marks each)

Q5. Explain the difference between crystalline and amorphous solids with two examples each.

[2 marks]

CRYSTALLINE SOLIDS:

- Have definite geometric shape and regular arrangement of particles
- Sharp melting point
- True solids, anisotropic

Student Name: Priyanka Nair

Roll No: 12CHE003

Subject: Chemistry

- Examples: NaCl, Diamond, Quartz, Sugar

AMORPHOUS SOLIDS:

- No definite shape, irregular arrangement
- Melt over a range of temperature
- Pseudo-solids or supercooled liquids, isotropic
- Examples: Glass, Rubber, Plastic, Tar

Key difference: Crystalline have long range order; amorphous have only short range order.

Marks	
Awarded	
/2	
Initials	

Q6. What are colloids? Give the classification based on physical state of dispersed phase and medium.

[2 marks]

Colloids (or colloidal solutions) are heterogeneous systems where dispersed phase particle size is between 1nm to 1000nm (1 μ m).

They do not settle on their own and cannot pass through semipermeable membranes.

Classification:

- Sol: solid in liquid (paint, mud)
- Gel: liquid in solid (jelly, cheese)
- Emulsion: liquid in liquid (milk, cream)
- Foam: gas in liquid (shaving cream)
- Aerosol: liquid in gas (fog, mist)
- Solid aerosol: solid in gas (smoke, dust)

Marks	
Awarded	
/2	

Student Name: Priyanka Nair

Roll No: 12CHE003

Subject: Chemistry

Initials	
----------	--

Q7. Write the IUPAC names of the following: (a) CH_3CHO (b) $\text{C}_6\text{H}_5\text{COCH}_3$

[2 marks]

(a) CH_3CHO :

IUPAC Name: Ethanal

The parent chain has 2 carbons with CHO (aldehyde) group at C-1.

Common name: Acetaldehyde

(b) $\text{C}_6\text{H}_5\text{COCH}_3$:

IUPAC Name: 1-Phenylethan-1-one (or Acetophenone by common name)

It is an aryl ketone with benzene ring attached to carbonyl group.

Alternate IUPAC: Methyl phenyl ketone

Marks	
Awarded	
/2	
Initials	

SECTION C — Long Answer I (4 marks each)

Q8. Discuss the preparation of ethanol by fermentation. Write the chemical equations involved.

[4 marks]

FERMENTATION OF SUGARS TO ETHANOL:

Step 1: Starch hydrolysis (if starting from starch)

Starch $\xrightarrow{\text{Diastase enzyme}}$ Maltose ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$)

Step 2: Maltose hydrolysis

 $\text{C}_{12}\text{H}_{22}\text{O}_{11} + \text{H}_2\text{O} \xrightarrow{\text{Maltase}}$ $2\text{C}_6\text{H}_{12}\text{O}_6$ (Glucose)

Step 3: Fermentation

 $\text{C}_6\text{H}_{12}\text{O}_6 \xrightarrow{\text{Zymase/Yeast}}$ $2\text{C}_2\text{H}_5\text{OH} + 2\text{CO}_2$

(Glucose) (Ethanol) (Carbon dioxide)

Conditions: Temperature $25\text{--}35^\circ\text{C}$, anaerobic conditions, pH 4-5

Purification: The mixture contains 15-18% ethanol. Distillation gives 95% ethanol (rectified spirit). Further dehydration over quick lime gives absolute alcohol (99.9%).

Note: CO_2 gas evolution can be detected by passing through lime water (turns milky).

Student Name: Priyanka Nair

Roll No: 12CHE003

Subject: Chemistry

Marks	
Awarded	
/4	
Initials	

SECTION D — Long Answer II (5 marks each)

Q9. Describe the extraction of copper from its ore. Give all reactions.

[5 marks]

ORE: Copper pyrites (CuFeS_2) is the main ore.

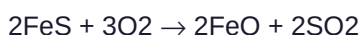
STEP 1 — CONCENTRATION:

Froth flotation method. Pine oil added to crushed ore + water.

Air blown through mixture. Cu ore particles attach to froth and float.

Gangue particles settle down. Froth removed and dried.

STEP 2 — ROASTING:

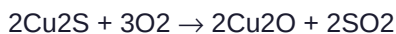
Concentrated ore heated in excess air at $700-800^\circ\text{C}$:

STEP 3 — SMELTING:

Roasted ore melted with silica (SiO_2) as flux:Remaining: $\text{Cu}_2\text{S} + \text{FeS}$ (matte)

STEP 4 — BESSEMERISATION:

Matte heated in Bessemer converter with air:



STEP 5 — ELECTROLYTIC REFINING:

Blister copper = anode, Pure copper = cathode

Electrolyte: $\text{CuSO}_4 + \text{H}_2\text{SO}_4$ solution

Cu dissolves from anode, deposits on cathode (99.9% pure)

Noble metals (Au, Ag) collect as anode mud.

Student Name: Priyanka Nair

Roll No: 12CHE003

Subject: Chemistry

Marks	
Awarded	
/5	
Initials	

Q10. Explain the mechanism of SN1 and SN2 reactions with suitable examples. Compare their characteristics.

[5 marks]

SN1 REACTION (Unimolecular Nucleophilic Substitution):

Mechanism (2 steps):

Step 1 (slow, rate-determining): $R-X \rightarrow R^{\bullet} + X^{\bullet}$ (carbocation formation)

Step 2 (fast): $R^{\bullet} + Nu^{\bullet} \rightarrow R-Nu$

Example: $(CH_3)_3C-Br + OH^{\bullet} \rightarrow (CH_3)_3C-OH + Br^{\bullet}$

Rate = $k[(CH_3)_3CBr]$ (first order, depends only on substrate)

Characteristics: Favoured by 3° alkyl halides, polar protic solvents, racemisation occurs

SN2 REACTION (Bimolecular Nucleophilic Substitution):

Mechanism (1 step, concerted):

$Nu^{\bullet} + R-X \rightarrow [Nu\cdots R\cdots X]^{\ddagger} \rightarrow Nu-R + X^{\bullet}$

(transition state)

Example: $CH_3Br + OH^{\bullet} \rightarrow CH_3OH + Br^{\bullet}$

Rate = $k[CH_3Br][OH^{\bullet}]$ (second order)

Characteristics: Favoured by 1° alkyl halides, polar aprotic solvents, inversion of configuration (Walden inversion)

COMPARISON TABLE:

1° substrate: SN2 faster

3° substrate: SN1 faster

Stereo: SN1 = racemic; SN2 = inversion

Solvent: SN1 = protic; SN2 = aprotic

Subject: Chemistry

Marks	
Awarded	
/5	
Initials	

Student Name: Priyanka Nair Roll No: 12CHE003 Subject: Chemistry

*** END OF ANSWER BOOKLET ***

MARKS SUMMARY (For Examiner Use Only)

Section	Max Marks	Marks Obtained	Examiner Initials
Section A (1 mark × 4)	4		
Section B (2 marks × 3)	6		
Section C (4-5 marks)	varies		
Section D (5-6 marks)	varies		
TOTAL	100		